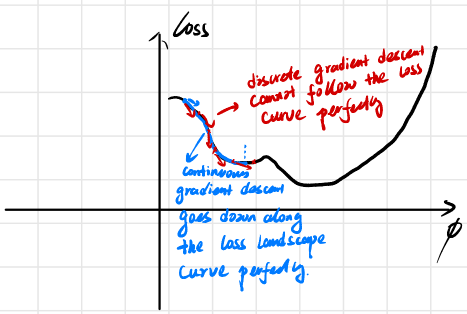
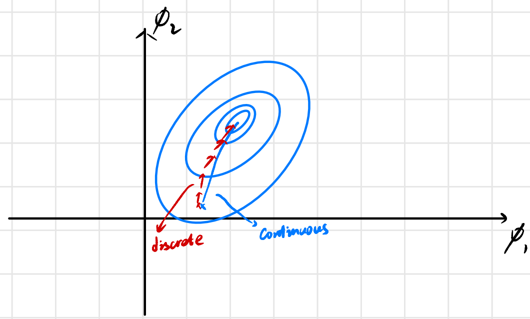
Suppose we have a model parametrized by ϕ, and we have a loss function $L(\phi)$. If we minimze this loss function using gradient descent or Stochastic gradient descent (SGD), we cannot optimize the loss function $L(\phi)$ directly because the discrete step size. On the contrary, imagine that the step size can be infinitesimal, then the loss will decrease with loss function loss scope perfectly.

→ we cannot physically do this

So the gradient descent method is minimizing $L(\phi)$ directly

if step size $\eta \to 0$, or if it always pass the point as continuous gradient descent.





Discrete Gradient Descent

$$\phi_1 = \phi_0 - \eta \nabla L(\phi_0)$$

Continuous Gradient Descent (in order words, the step size is infinitesimal)

instead of discrete steps, we imagine the parameters flowing continuously over time t, at each point, we have

$$\frac{d\phi}{dt} = -\nabla_{\phi(t)} L(\phi(t))$$

→ assume ϕ follows over time

Gradient descent $\nearrow$ does not optimize the loss function $L(\phi)$ directly

$\searrow$ assume it is optimizing a slightly different loss function $\tilde{L}(\phi)$

implication here: the discrete gradient descent and continuous gradient descent will work exactly the same in this modified loss $\tilde{L}(\phi)$, i.e.

$$-\eta \nabla \tilde{L}(\phi_0) = \int_0^{\eta} \nabla L(\phi(t))\, dt$$

assume $\phi(0) = \phi_0$

goal: find the modified loss $\tilde{L}(\phi)$ (the loss where gradient descent optimize directly)

Let us assume the gradient of this modified loss equals

$$\frac{d\tilde{L}(\phi(t))}{dt} = \left(\underbrace{-\frac{dL(\phi)}{d\phi}}_{\text{original gradient}} + \eta \underbrace{g(\phi)}_{\text{correction term}} \right) \Bigg|_{\phi=\phi(t)} \qquad (2)$$

Then Taylor expansion of $\phi(t+\eta)$ around $\phi(t)$ gives

here $\phi(t)$ is a function of time t where at each point we have $\frac{d\phi}{dt} = \nabla \tilde{L}(\phi(t))\Big|_{\phi(t)}$

$$\phi(t+\eta) = \left(\phi(t) + \eta \frac{d\phi(t)}{dt} + \frac{\eta^2}{2} \frac{d^2\phi(t)}{dt^2} + O(\eta^3) \right) \Bigg|$$

t is a constant such that $\phi(t) = \phi_0$

drop $O(\eta^3)$: the Lagrange Remainder

$$\approx \phi_0 + \eta \nabla \tilde{L}(\phi_0) + \frac{\eta^2}{2} \nabla^2 \tilde{L}(\phi_0)$$

$$= \phi_0 + \eta(-\nabla L(\phi_0) + \eta g(\phi_0)) + \frac{\eta^2}{2}\left(-\frac{d\nabla L(\phi)}{d\phi} \frac{d\phi(t)}{dt} + \eta \frac{dg(\phi)}{d\phi} \frac{d\phi(t)}{dt} \right)$$

$$= \phi_0 + \eta(-\nabla L(\phi_0) + \eta g(\phi_0)) + \frac{\eta^2}{2} \frac{d\nabla L(\phi)}{d\phi} \nabla L(\phi_0) + \frac{\eta^3}{2} \frac{dg(\phi)}{d\phi} \frac{d\phi(t)}{dt}$$

drop $O(\eta^3)$ again

$$\approx \phi_0 - \eta \nabla L(\phi_0) + \eta^2 \left(g(\phi_0) + \frac{1}{2} \frac{d\nabla L(\phi)}{d\phi} \nabla L(\phi_0) \right)$$

note that $\phi_0 - \eta\nabla L(\phi_0)$ is the updated the discrete gradient descent make, So in order to make discrete and continuous gradient descent to get to the same place, we must have

$$g(\phi_0) + \frac{1}{2}\frac{d\nabla L(\phi)}{d\phi}\nabla L(\phi_0) = 0 \quad \Rightarrow \quad g(\phi_0) = -\frac{1}{2}\underbrace{\frac{d^2 L(\phi)}{d\phi^2}}_{\text{Hessian matrix}}\nabla L(\phi_0)$$

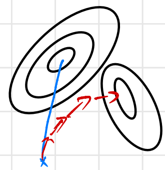
$$\overset{(2)}{\Rightarrow} \quad \frac{d\tilde{L}(\phi)}{dt} \approx -\frac{dL(\phi)}{d\phi} - \frac{1}{2}\eta\frac{d^2 L(\phi)}{d\phi^2}\nabla L(\phi_0)$$

$$\overset{\text{matrix calculus}}{\Rightarrow} \quad \tilde{L}(\phi) \approx L(\phi) + \frac{1}{4}\eta\left\|\frac{\partial L}{\partial \phi}\right\|^2 \quad ☆$$

the minimum position of the continuous and discrete gradient descent is the same.

but the path differ. So they may end up different minimum

Implications: the discrete gradient descent perfers the place where the gradient norm is smaller.



Implicit regularization in SGD

using the similar analysis, we can show that

$$\tilde{L}_{SGD} = L(\phi) + \frac{1}{4}\eta\left\|\frac{\partial L}{\partial \phi}\right\|^2 + \frac{1}{4}\eta \underbrace{B}_{\text{total number of batches}} \underbrace{\sum_{b=1}^{B}\left\|\frac{\partial L_b}{\partial \phi} - \frac{\partial L}{\partial \phi}\right\|^2}_{\text{extra regularization}}$$

Compared with gradient descent, SGD perfers the position where the gradient is stable

(the gradients for each batch agrees with the total gradient)